

Overview of ASP.Net Core MVC

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- ◆ ASP.NET Core is a cross-platform, high-performance, open-source framework for building modern, cloud-enabled, Internet-connected apps.
- ◆ With ASP.NET Core, you can:
 - ◆ Build web apps and services, Internet of Things (IoT) apps, and mobile backends.
 - ◆ Use your favorite development tools on Windows, macOS, and Linux.
 - ◆ Deploy to the cloud or on-premises.
 - ◆ Run on .NET Core.

ASP.NET Core provides the following benefits

- ◆ A unified story for building web UI and web APIs.
- ◆ Architected for testability.
- ◆ Razor Pages makes coding page-focused scenarios easier and more productive.
- ◆ Blazor lets you use C# in the browser alongside JavaScript. Share server-side and client-side app logic all written with .NET.
- ◆ Ability to develop and run on Windows, macOS, and Linux.
- ◆ Open-source and community-focused .
- ◆ Integration of modern, client-side frameworks and development workflows.
- ◆ Support for hosting Remote Procedure Call (RPC) services using gRPC.
- ◆ A cloud-ready, environment-based configuration system.
- ◆ Built-in dependency injection.
- ◆ A lightweight, high-performance , and modular HTTP request pipeline.

Build web APIs and web UI using ASP.NET Core MVC

- ◆ The Model-View-Controller (MVC) pattern helps make your web APIs and web apps testable.
- ◆ Razor Pages is a page-based programming model that makes building web UI easier and more productive.
- ◆ Razor markup provides a productive syntax for Razor Pages and MVC views.
- ◆ Tag Helpers enable server-side code to participate in creating and rendering HTML elements in Razor files.
- ◆ Built-in support for multiple data formats and content negotiation lets your web
- ◆ APIs reach a broad range of clients, including browsers and mobile devices.
- ◆ Model binding automatically maps data from HTTP requests to action method parameters.
- ◆ Model validation automatically performs client-side and server-side validation.

MVC Pattern

- ◆ MVC Pattern

- ◆ Model

- ◆ View

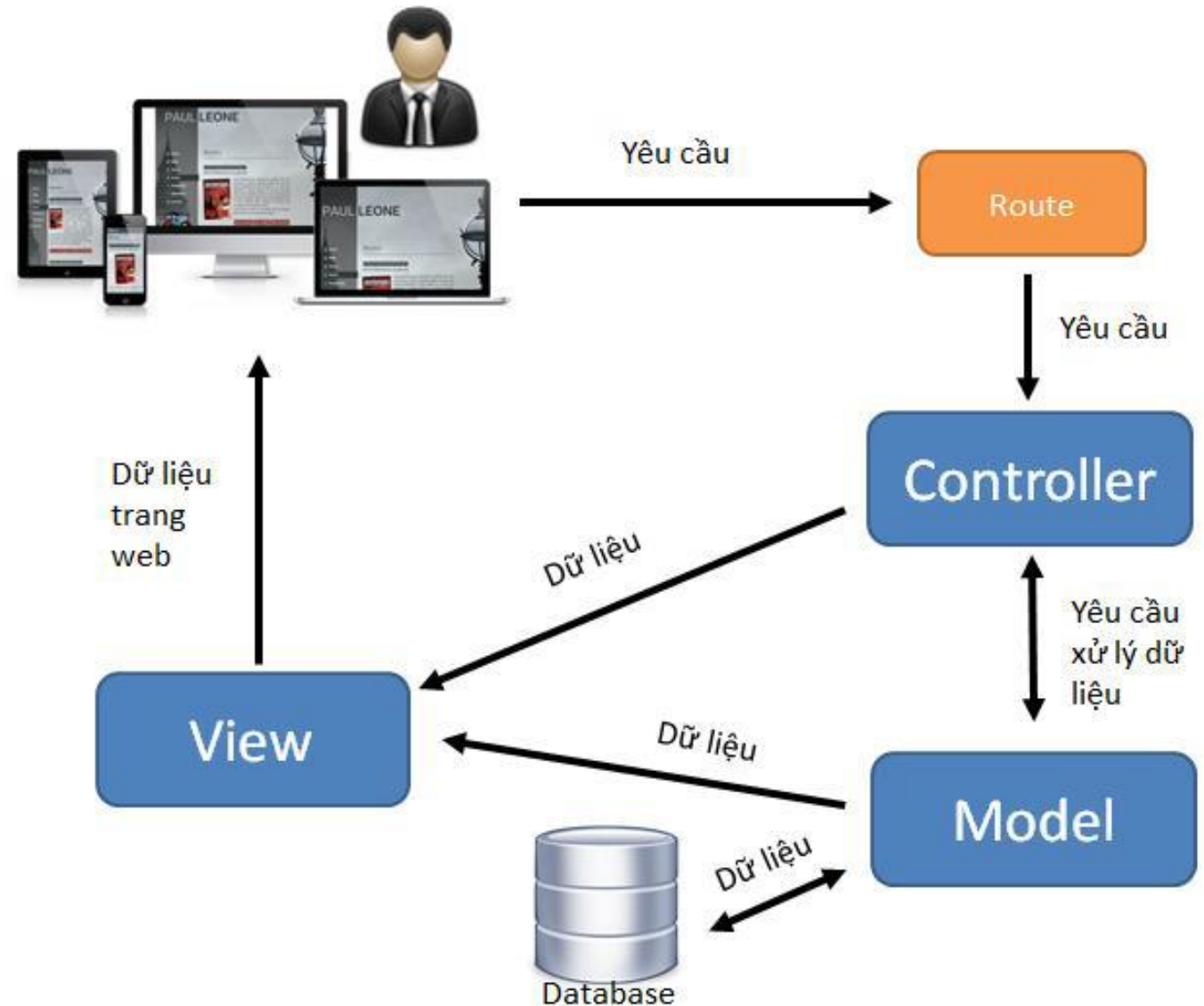
- ◆ Controller

- ◆ MVC Framework

- ◆ Java: Struts, Spring, JSF...

- ◆ PHP: Zend

- ◆ ASP.Net MVC



Create a web app

- ◆ Start Visual Studio and select Create a new project.
- ◆ In the Create a new project dialog, select ASP.NET Core Web App (Model- View-Controller) > Next.
- ◆ In the Configure your new project dialog, enter MvcMovie for Project name. It's important to name the project MvcMovie. Capitalization needs to match each namespace when code is copied.
- ◆ Select Next.
- ◆ In the Additional information dialog, select .NET 6.0 (Long-term support).
- ◆ Select Create.

Create a web app

Additional information

ASP.NET Core Web App (Model-View-Controller) C# Linux macOS Windows Cloud Service Web

Framework ⓘ
.NET 6.0 (Long-term support) ▼

Authentication type ⓘ
None ▼

☒ Configure for HTTPS ⓘ

☐ Enable Docker ⓘ

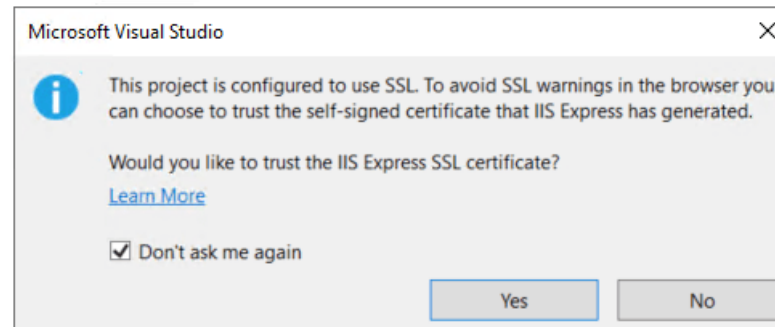
Docker OS ⓘ
Linux ▼

Back Create

Run the app

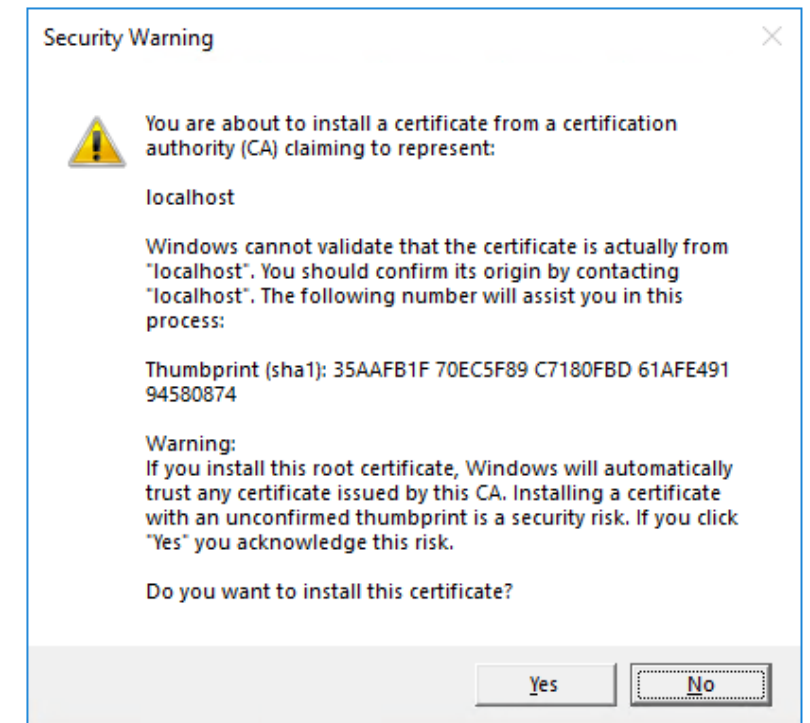
- ◆ Select Ctrl+F5 to run the app without the debugger.

Visual Studio displays the following dialog when a project is not yet configured to use SSL:



- ◆ Select Yes if you trust the IIS Express SSL certificate.

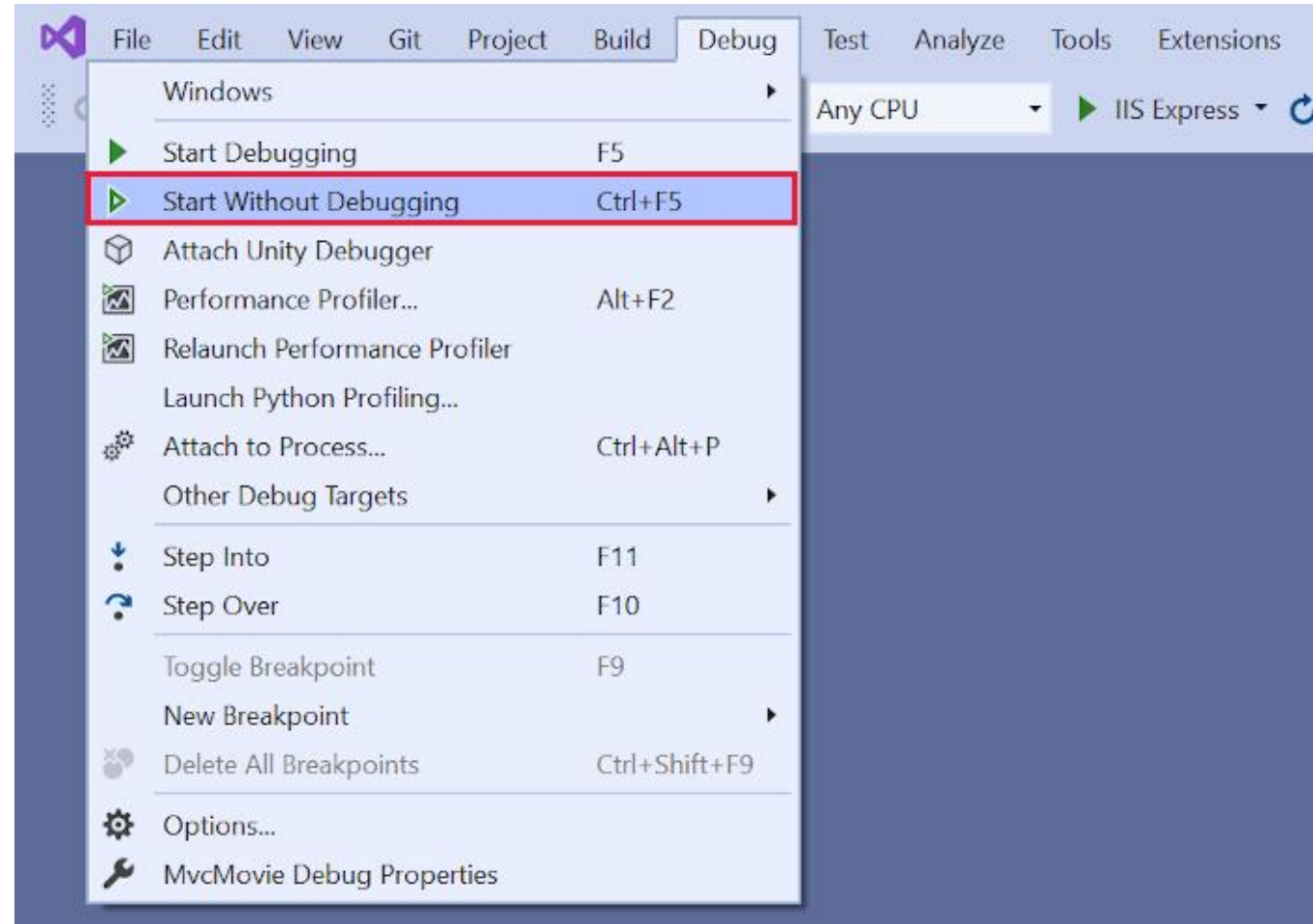
The following dialog is displayed



- ◆ Select Yes if you agree to trust the development certificate.

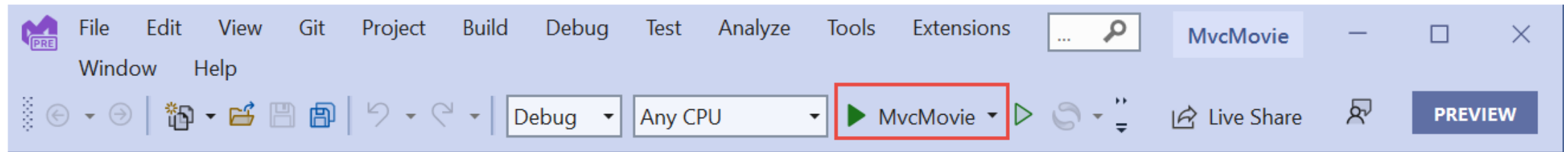
Launching the app without debugging

- ◆ by selecting Ctrl+F5 allows you to
 - ◆ Make code changes.
 - ◆ Save the file.
 - ◆ Quickly refresh the browser and see the code changes.



Run the app

- ◆ You can debug the app by selecting the MvcMovie button in the toolbar:



Run the app

